Ashsmith Khayrul

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Education

Northeastern University Sep. 2025 – Dec. 2028
B.S. in Computer Science and Business Administration *Boston, MA*
GPA: 3.64/4.0 (Overall), 3.94/4.0 (CS), *Dean's List*
Relevant Coursework: Artificial Intelligence, Object-Oriented Design, Databases, Programming Languages, Discrete Mathematics, Business Statistics, Finance, Marketing, Microeconomics, Accounting
Activities: NU Entrepreneurs Club (Director of Technology)

Technical Skills

Languages: Python, Java, JavaScript, Pyret, SQL (MySQL), HTML/CSS, PHP
Technologies: React, Node.js, Express, FastAPI, REST APIs, Redis, MongoDB, WordPress
Developer Tools: Git, GitHub Actions, Docker, Claude Code, Codex, CI/CD, Shell, Nginx, VS Code, Visual Studio
Systems: Linux, macOS, Windows
Certifications: Bloomberg Market Concepts (Issued Aug. 2026)

Experience

Northeastern University Sept. 2026 – Present
Bloomberg Lab Manager *Boston, MA*
- Manage and maintain **20 Bloomberg Terminal stations** for the D'Amore-McKim Bloomberg Business Lab, performing daily opening and closing procedures and monitoring equipment usage.
- Coordinate with university IT to troubleshoot technical issues across lab workstations, ensuring consistent access to real-time market data and financial analytics tools.
- Train and support students on Bloomberg Terminal navigation and financial data workflows, leading orientation sessions for new lab users.

Northeastern University May 2026 – Present
Undergraduate Teaching Assistant *Boston, MA*
- Served as a Teaching Assistant for CS 3200: Database Design (Summer 2026) and CS 2000: Introduction to Program Design and Implementation (Fall 2026).
- Held office hours for **110+ students** in CS 3200, assisting with SQL queries, schema design, normalization, and relational database concepts, and graded ER modeling and MySQL implementations with written feedback.
- Supported CS 2000 students with program design, debugging, data analysis, and programming in Pyret and Python.

Uniplex Host Apr. 2024 – Present
Founder & Systems Engineer *Remote*
- Built and operated a Docker-based hosting platform, serving **40+ clients** across **65+ active services**, offering VPS, game servers, and bot hosting across U.S. nodes, with automated provisioning and billing integrations via WHMCS APIs enabling scalable service lifecycle management.
- Maintained **99.9% uptime** across all production services with **24/7 SLA-backed containers**, supported by Nginx reverse proxy configuration, real-time status monitoring, and automated recovery mechanisms.
- Managed multi-node Linux infrastructure with optimized compute and memory allocation, operating a full service suite including a game panel, web panel, status page, and documentation portal.

Northeastern University Jan. 2026 – Apr. 2026
Undergraduate Research Assistant *Oakland, CA*
- Supported a City of Oakland–Northeastern University partnership evaluating responsible AI adoption in local government, focusing on Microsoft Copilot's effectiveness and risks in scaling community support services.
- Analyzed **1,000+ data points** from IRB-approved surveys and usage logs to assess how prompt engineering skills evolve with AI tool exposure and what drives user trust vs. over-reliance.

Projects

AshBot | *Discord API, JavaScript, Node.js, MySQL, Redis* July 2023 – Feb. 2026
- Architected and deployed a Node.js backend serving **1,200+ Discord servers** and **250,000+ users**, implementing event-driven asynchronous command handling for high-concurrency workloads.
- Designed an indexed MySQL data layer with Redis caching to reduce database load and improve response latency under peak traffic; migrated from a single-process deployment to a sharded, containerized Docker architecture on Linux VPS infrastructure.

StreamCompass | *Python, FastAPI, pandas, NumPy, TF-IDF* May 2026 – June 2026
- Built the ML/backend layer for a content-based recommendation engine over **42,000+ streaming titles**, using weighted TF-IDF and cosine similarity across descriptions, genres, cast, and director metadata to rank movie and TV recommendations.
- Designed the FastAPI recommendation API, data loading pipeline, platform/IMDb filters, and match-breakdown response fields that powered frontend search, filtering, and recommendation explanations.